

Two Pointers Level -2 Problems

1. Two Sum II — Input Array Is Sorted Pattern: Opposite Two Pointers

```
nums = [2,7,11,15]
target = 9

def two_sum(nums, target):

    left = 0
    right = len(nums) - 1

    while left < right:

        total = nums[left] + nums[right]

        if total == target:
            return [left, right]

        elif total < target:
            left += 1

        else:
            right -= 1

    return []

print(two_sum(nums,target))
```

3. Move Zeroes Pattern: Slow + Fast Pointer

```
nums = [0,1,0,3,12]

def move_zeroes(nums):

    slow = 0

    for fast in range(len(nums)):

        if nums[fast] != 0:
            nums[slow],nums[fast] = nums[fast], nums[slow]

            slow +=1

    return nums

print(move_zeroes(nums))
```

5. Squares of a Sorted Array Pattern: Left + Right

```
nums = [-4, -1, 0, 3, 10]

def sorted_sqr(nums):

    n = len(nums)
    result = [0]*n

    left = 0
    right = n -1

    for position in range(n-1,-1,-1):

        if abs(nums[left]) > abs(nums[right]):
            result[position] = nums[left] **2
            left +=1

        else:
            result[position] = nums[right] **2
            right -=1

    return result

print(sorted_sqr(nums))
```

2. Remove Duplicates from Sorted Array Pattern: Slow + Fast Pointer

```
nums = [1,1,2,2,3]

def remove_dup(nums):

    if not nums:
        return 0

    slow = 0

    for fast in range(1, len(nums)):

        if nums[fast] != nums[slow]:
            slow +=1

        nums[slow] = nums[fast]

    return slow + 1

print(remove_dup(nums))
```

4. Remove Element Pattern: Slow + Fast

```
nums = [3,2,2,3]
target = 3

def remove_element(nums, target):

    slow = 0

    for fast in range(len(nums)):

        if nums[fast] != target:
            nums[slow] = nums[fast]
            slow += 1

    return slow

print(remove_element(nums,target))
```

6. Merge Sorted Arrays Pattern: Two Pointer Merge

```
nums1 = [1,2,3,0,0,0]
m = 3
nums2 = [2,5,6]
n = 3

def merge(nums1, m, nums2, n):

    i = m - 1
    j = n - 1
    k = m + n - 1

    while i >= 0 and j >= 0:
        if nums1[i] > nums2[j]:
            nums1[k] = nums1[i]
            i -= 1
        else:
            nums1[k] = nums2[j]
            j -= 1
        k -= 1

    while j >= 0:
        nums1[k] = nums2[j]
        j -= 1
        k -= 1

    return nums1

print(merge(nums1,m,nums2,n))
```